Michael Ji

647-938-0623 | m2ji@uwaterloo.ca | linkedin.com/in/michael-jw-ji | github.com/michael-jw-ji | michael-ji.vercel.app

Education

University of Waterloo — Waterloo, ON
Bachelor of Computing and Financial Management (Honors, Co-op) — *Expected April 2028*

Experience

Software Engineer Intern — September 2026 – Present
Communitech (Data & AI Automation @ Langdon Hall) — *Waterloo, ON*

- Building **Python/SQL** ingestion and aggregation pipelines on **Microsoft Fabric**, transforming reservation, spa, restaurant, and POS data from **4 source systems** into a unified guest and revenue model for cross-department analytics.
- Built a **Claude-powered** report generator on top of the unified guest model, turning data from 4 systems into daily and per-guest department summaries via a curated system prompt, cutting lookup time from **30 to 2 minutes**.
- Hardening the platform for guest PII with row-level security, least-privilege **Entra ID** service principals per pipeline, secrets in **Azure Key Vault**, and Canadian-region data residency for **PIPEDA** compliance.

Software Engineer Intern — May 2026 – August 2026
Enns Real Estate (Full-Stack Deployment) — *Waterloo, ON*

- Designed and built the brokerage's listings platform end to end on **Next.js/TypeScript** and **Supabase**, from schema design to deployment, contributing to **200% year-over-year revenue growth** after launch with 200+ monthly visitors.
- Reduced listing page load time by **40%** by caching listing queries with **Next.js** incremental static regeneration and indexing Postgres filter columns for price, bedrooms, and location, revalidating pages on every listing update.
- Automated staging and production releases with **GitHub Actions** and **Vercel CI/CD**, gating every PR on typecheck, lint, and build checks, cutting release time to **under 5 minutes** with **0 failed production deploys.**

Software Engineer Intern — January 2026 – April 2026
Rogers Communications (AI Governance Platform) — *Toronto, ON*

- Built **Python** APIs and document ingestion pipelines for an internal policy assistant serving **5K+ users**, indexing 500+ policy documents into **Azure AI Search**, handling **35K+ weekly queries** at 3s p95 latency with **Azure OpenAI**.
- Designed request-validation middleware in front of the RAG APIs with LLM-based query normalization, prompt-injection detection, oversized-input rejection, and file-type allowlisting, blocking **95%** of 250+ adversarial test cases.
- Shipped an async pipeline alerting owners on suspicious prompts via Azure OpenAI function calling and Power Automate, decoupled from the response path to keep user **p95 latency at 3s**, cutting review time from **2 hours to 5 minutes**.

Machine Learning Research Intern — April 2025 – March 2026
Algoverse (Model Interpretability) — *Palo Alto, CA*

- Built a Python pipeline labeling 5 variable roles across 7 programming languages with AST and regex extractors, using problem-hash 70/10/20 splits to prevent train/test leakage; released as a public Hugging Face dataset.
- Built a model-agnostic CLI extracting per-layer hidden states from any Hugging Face LLM and trained probes reaching 0.98 macro-F1 across 3 models, using perturbation and random-label controls to show high scores can reflect lexical shortcuts.

Projects

LLM Filing Extraction and Evaluation Platform | *Python, Docker, OpenTelemetry, pytest, Postgres*

- Raised field-level accuracy from **84% to 99.3%** and valid output from **55% to 99.5%** across 189 SEC insider filings with a Python pipeline that extracts via LLM and validates each filing against **schema and business rules**.
- Built a **GitHub Actions release gate** that replays recorded model outputs on every PR and blocks merges on **statistically significant per-field regressions**, catching a bug that passed **all 75 unit tests**.
- Made every code change re-scorable with zero new model calls via **content-addressed caching** of LLM outputs, plus retries with backoff, a rate-limited SEC fetcher, and **OpenTelemetry tracing** on every model call (7.6s p50, 63s p99).

Serverless Filing Ingestion and Query Platform | *Python, AWS Lambda, EventBridge, SQS, DynamoDB*

- Built an event-driven AWS pipeline ingesting **10K+ insider filings** with idempotent writes and **0 duplicates**.
- Designed a DynamoDB schema with composite keys serving per-insider history at **p95 reads under 300ms**.

Publications

Apparent Cross-Language Invariance Reflects Shared Identifiers in Parallel Programs. N. O. Lwin, S. D. Nimmagadda, A. Dutt, **M. Ji**, et al. *NeurIPS 2026 Workshop (LP4FM)*

Same Score, Different Evidence: Decodability, Surface Sufficiency, and Causal Relevance in Code Models. N. O. Lwin, A. Dutt, S. D. Nimmagadda, **M. Ji**, et al. *NeurIPS 2026 Workshop (InterpScience)*

Skills

Languages: Python, C/C++, TypeScript/JavaScript, SQL, Java, Go, Rust
Databases & Frameworks: PostgreSQL, DynamoDB, Supabase, pytest, FastAPI, Flask, Next.js, React.js, Node.js
Technologies: AWS, Azure, Docker, GitHub Actions, Vercel, Git, REST APIs, Linux, Kubernetes, Kafka